Kaveh Dastouri

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Education

PhD Tarbiat Modares University, Tehran, Iran 2020 – 2023
Field: Algebra
Thesis: "Monomial characters and structure of finite groups"
Supervisor: Prof. Ali Iranmanesh
Advisor: Prof. Mohsen Ghasemi

MS University of Tabriz, Tabriz, Iran 2017 – 2020
Field: Algebra
Thesis: "Irreducible characters of even degree and normal Sylow 2-subgroups"
Supervisor: Prof. Kamal Aziziheris

BS Farhangian University, Iran 2013 – 2017
Field: Mathematics Education

Research Experience

Research Stay RPTU Kaiserslautern-Landau, Germany Jul 2026 – Dec 2026
Host: Prof. Günter Malle.

Postdoc Beijing International Center for Mathematical Research (BICMR), Peking University Dec 2025 – present
Supervisor: Prof. Jiping Zhang

Postdoc Department of Pure Mathematics, Tarbiat Modares University Jun 2024 – Sep 2025
Supervisor: Prof. Ali Iranmanesh

Honors and Awards

DAAD Research Grant (German Academic Exchange Service) 2026
Competitive research scholarship for a six-month research in Germany.

Highest GPA among PhD Students 2024
Highest grade point average among algebra PhD students at Tarbiat Modares University.

Qualified as an "Exceptional Talent" PhD Student 2020
Exempt from the entrance exam, Tarbiat Modares University.

Publications

Journal Papers

[1] Kaveh Dastouri. *Symmetric Group-Based Public-Key Cryptosystem with Large Prime Moduli*. Cryptology ePrint Archive, Paper 2025/1589. 2025. URL: https://eprint.iacr.org/2025/1589.

[2] Kaveh Dastouri and Mohsen Ghasemi. "Groups with prime power Isaacs π-partial character codegrees." (Journal of Algebra and Its Applications). 2025. URL: https://doi.org/10.1142/S0219498827500897.

[3] Kaveh Dastouri and Ali Iranmanesh. "Primes and degrees of Isaacs π-partial characters." In: *Communications in Algebra* 53.6 (2025), pp. 2523–2527. DOI: 10.1080/00927872.2024.2446543. eprint: https://doi.org/10.1080/00927872.2024.2446543. URL: https://doi.org/10.1080/00927872.2024.2446543.

[4] Kaveh Dastouri, Ali Iranmanesh, and Mohsen Ghasemi. "Monomial Isaacs π-partial characters and derived length." In: *Journal of Algebra and Its Applications* 24.04 (2025), p. 2550108. DOI: 10.1142/S0219498825501087. eprint: https://doi.org/10.1142/S0219498825501087. URL: https://doi.org/10.1142/S0219498825501087.

[5] Kaveh Dastouri, Ali Iranmanesh, and Mohsen Ghasemi. "Primes and Degrees of Isaacs π-partial Character II." (Submitted). 2025.

Preprints

1. K. Dastouri, "A Symmetric Group-Based Public-Key Cryptosystem with Hard-to-Factor One-Way Function," (submitted).
2. K. Dastouri and A. Iranmanesh, "Some Combinatorial Applications of Symmetric Group Representations," (submitted).
3. K. Dastouri, "Reconstruction of Permutation Sets from Their Decks Up to Conjugation,".
4. K. Dastouri and A. Iranmanesh, "Permutations of the Symmetric Group and Hamiltonian Cycles," (submitted).

Conference Papers

1. K. Dastouri, A. Molkhasi, "Pseudocomplements in free groups", The 4th Seminar on Algebra and its Applications, August 9-11, 2016.
2. K. Dastouri, A. Iranmanesh, "Primes and degrees of Isaacs π-partial character", 15th International Group Theory Conference of Iran, February 9-10, 2023.
3. K. Dastouri, A. Iranmanesh, "Some results of Isaacs π-partial character degrees", 4th International Conference on Computational Algebra, Computational Number Theory and Applications, July 4-6, 2023.
4. K. Dastouri, A. Iranmanesh, "On monomial Isaacs π-partial characters of π-separable groups", 7th Biennial International Group Theory Conference, August 7-11, 2023.
5. K. Dastouri, A. Iranmanesh, M. Ghasemi, "Groups whose all Isaacs π-partial characters are monomial", 16th International Group Theory Conference of Iran, February 1-2, 2024.
6. K. Dastouri, A. Iranmanesh, "Representation theory of symmetric groups and some combinatorial results", 12th Graph Theory and Algebraic Combinatorics, February 7-8, 2024.
7. K. Dastouri, A. Iranmanesh, "برخی نتایج در مورد نتایج درجات سرشت هاي مونولتیک", 55th Annual Iranian Mathematics Conference, August 14-16, 2024.
8. A. Iranmanesh, K. Dastouri, "π-structure and π-characters", 17th International Group Theory Conference of Iran, January 29-30, 2025.

Research Interests

Representation Theory of Finite Groups: ordinary and modular cases
Finite Groups and Their Applications
Public-Key Cryptosystem Design: leveraging hard computational representation problems

Teaching Experience

Teaching Assistant	Tarbiat Modares University, Tehran, Iran *Courses:* Representation Theory of Finite Groups, Abstract Algebra Lecturer: Prof. Ali Iranmanesh	2020 – 2024
High School Teacher	Various High Schools, Iran Mathematics instructor for 7 years	2013 – 2020

Computer Skills

GAP	Advanced
SageMath	Intermediate
Python (NumPy, TensorFlow, Keras)	Familiar
LaTeX	Advanced
Microsoft Office	Advanced
Linux	Familiar

Languages

Persian	Native
English	Upper Intermediate (C1)
German	Basic (A1)

References

Prof. Ali Iranmanesh
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Prof. Jiping Zhang
School of Mathematical Sciences, Peking University, China
Email: `jzhang@pku.edu.cn`

Prof. Mohammad Reza Darafsheh
School of Mathematics, Statistics and Computer Science, University of Tehran, Iran
Email: `darafsheh@ut.ac.ir`

Dr. Ali Rajaei
Faculty of Mathematical Sciences, Tarbiat Modares University, Iran
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